

UCS660: Continuous Integration and Continuous Deployment

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Course Objective: The objective of the course is to teach techniques to automate the process of integration and deployment of software product. It covers prerequisites, anatomy and framework/tools used for the automated process of continuous integration and continuous deployment.

Syllabus

DevOps Automation: Phases in software development-delivery pipeline, components of automated software delivery, RAD model and model driven architecture.

Automation Benefits: advantages of automation, Time and efforts saving scenarios, error preventing scenarios.

Continuous Integration and Continuous Deployment Introduction: Overview and practices of continuous integration, working mechanism and benefits of continuous integration; continuous delivery's introduction and pipeline. Prerequisites and benefits, introduction and business drivers of continuous deployment, benefits of continuous deployment.

Stages and Anatomy of CI CD: Core continuous integration process and advanced continuous integration process, release process, continuous delivery engineering practices, continuous testing & promotion of builds, continuous monitoring of delivery pipeline, understanding continuous feedback process.

Testing, Debugging and Refactoring: Understanding test-driven development (TDD), categories of TDD, Junit framework, need for code refactoring, its process and strategies.

Understanding Framework and Tools: Common frameworks and code architectures, third party code, IDEs (Eclipse, Netbeans and IntelliJ), common mistakes and avoiding them, issues with making code IDE dependent.

Laboratory Work: Setting up Jenkins, Jenkins job, parameters, build, post-build actions and pipeline; Jenkins plugins, using Jenkins as a continuous integration server; Configuring Jenkins with git plugin; Jenkins pipeline to poll the feature branch.

Course Learning Objectives (CLO)

After completion of this course, the students will be able to:

1. Identify the need for migrating from traditional software development to Agile model and then to DevOps.
2. Define and understand the basic principles and need of DevOps and Continuous Delivery.
3. Understand the history and overview of Source Code Management, along with real-time examples.
4. Differentiate between centralized and distributed version control systems and basic operations in version control systems and Demonstrate the use of various version control systems.

Text Books

1. Gene Kim, Jez Humble, Patrick Debois, John Willis, "The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations", IT revolution Press (2016) 1st ed.

Reference Books:

1. Sander Rossel, “Continuous Integration, Delivery, and Deployment: Reliable and Faster Software Releases with Automating Builds, Tests, and Deployment”, Packt Publishing (2017) 1st ed.
2. Online material available at:<https://digitallearn.xebiaacademyglobal.com/>